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TASK-05:

CODE:

```
•[7]: import pandas as pd
      from sklearn.model_selection import train_test_split, cross_val_score
      from sklearn.tree import DecisionTreeClassifier, plot_tree
      from sklearn.ensemble import RandomForestClassifier
      from sklearn.metrics import accuracy_score, classification_report
      import matplotlib.pyplot as plt
      import seaborn as sns

      df = pd.read_csv(r"C:\Users\julug\Downloads\heart disease.csv")
      print("First 5 rows:\n", df.head())

      print("\nColumns:", df.columns.tolist())

      if 'target' not in df.columns:
          raise ValueError("Please replace 'target' with the correct column name.")

      df = df.dropna()

      X = df.drop(columns=["target"])
      y = df["target"]

      X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

      dt = DecisionTreeClassifier(max_depth=4, random_state=42)
      dt.fit(X_train, y_train)

      plt.figure(figsize=(20,10))
      plot_tree(dt, feature_names=X.columns, class_names=["No Disease", "Disease"], filled=True)
      plt.title("Decision Tree (max_depth=4)")
      plt.show()

      train_acc = dt.score(X_train, y_train)
      test_acc = dt.score(X_test, y_test)
      print(f"\nDecision Tree - Train Accuracy: {train_acc:.2f}")
      print(f"Decision Tree - Test Accuracy: {test_acc:.2f}")
```

Code

```
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
rf_pred = rf.predict(X_test)

print("\nRandom Forest - Test Accuracy:", accuracy_score(y_test, rf_pred))
print("\nClassification Report (Random Forest):\n", classification_report(y_test, rf_pred))

importances = pd.Series(rf.feature_importances_, index=X.columns)
importances.sort_values().plot(kind='barh', figsize=(10,6), color='skyblue')
plt.title("Feature Importances from Random Forest")
plt.xlabel("Importance")
plt.tight_layout()
plt.show()

dt_cv = cross_val_score(dt, X, y, cv=5)
rf_cv = cross_val_score(rf, X, y, cv=5)

print(f"\nDecision Tree CV Accuracy: {dt_cv.mean():.2f}")
print(f"Random Forest CV Accuracy: {rf_cv.mean():.2f}")
```

OUTPUT:

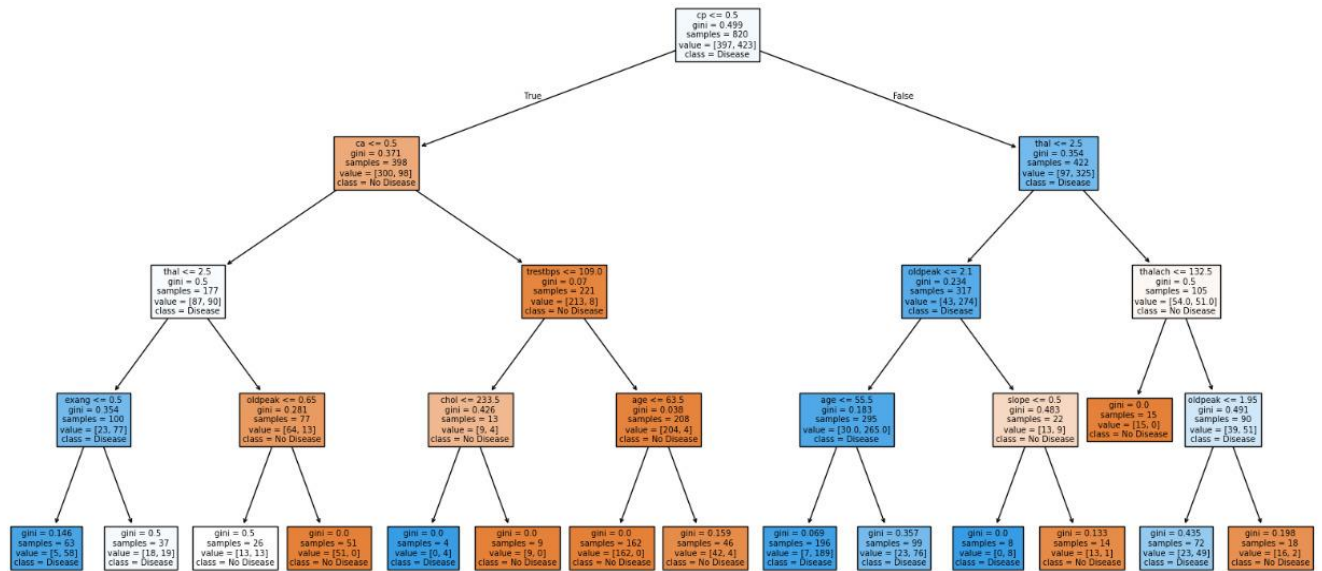
First 5 rows:

	age	sex	cp	trestbps	chol	fb	restecg	thalach	exang	oldpeak	slope	\
0	52	1	0	125	212	0	1	168	0	1.0	2	
1	53	1	0	140	203	1	0	155	1	3.1	0	
2	70	1	0	145	174	0	1	125	1	2.6	0	
3	61	1	0	148	203	0	1	161	0	0.0	2	
4	62	0	0	138	294	1	1	106	0	1.9	1	

	ca	thal	target
0	2	3	0
1	0	3	0
2	0	3	0
3	1	3	0
4	3	2	0

Columns: ['age', 'sex', 'cp', 'trestbps', 'chol', 'fb', 'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target']

Decision Tree (max_depth=4)



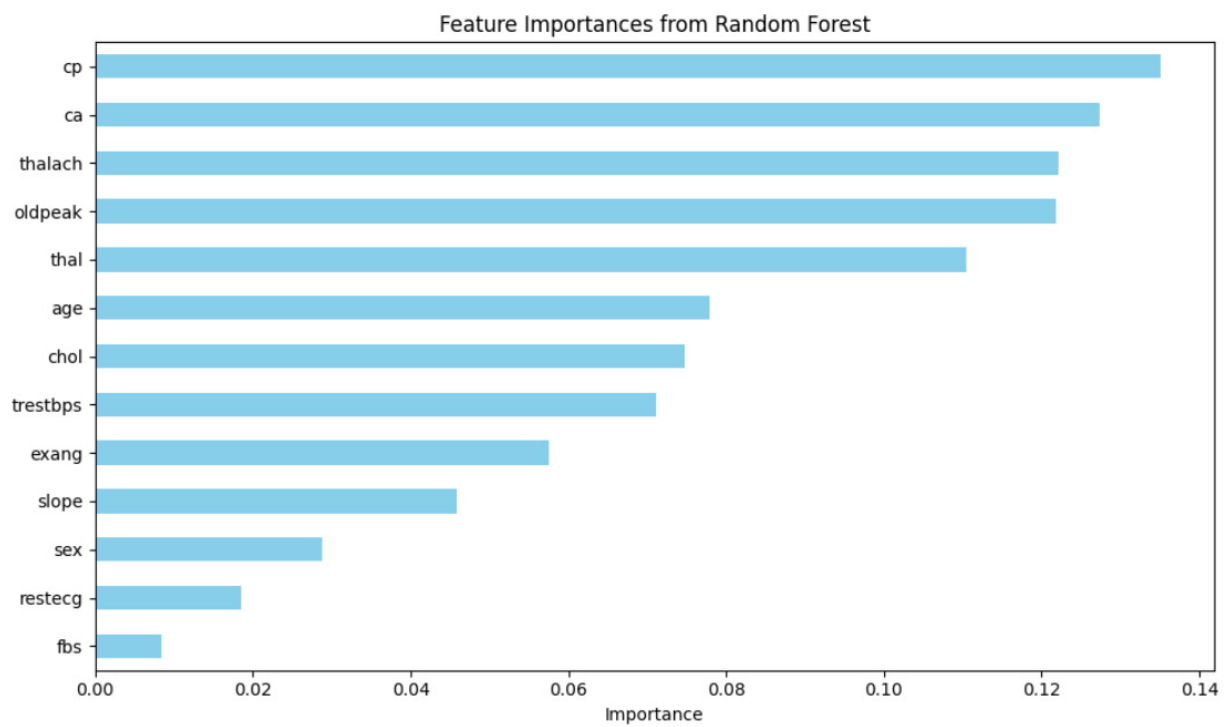
Decision Tree - Train Accuracy: 0.88

Decision Tree - Test Accuracy: 0.80

Random Forest - Test Accuracy: 0.9853658536585366

Classification Report (Random Forest):

	precision	recall	f1-score	support
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
accuracy			0.99	205
macro avg	0.99	0.99	0.99	205
weighted avg	0.99	0.99	0.99	205



Decision Tree CV Accuracy: 0.83

Random Forest CV Accuracy: 1.00